# National Testing Agency

| | |
|---|---|
| **Question Paper Name :** | B Tech 7 Apr 2025 Shift 2 |
| **Subject Name :** | B. Tech |
| **Duration :** | 180 |
| **Total Marks :** | 300 |
| **Display Marks:** | Yes |

## B. Tech

| | |
|---|---|
| **Group Number :** | 1 |
| **Group Maximum Duration :** | 0 |
| **Group Minimum Duration :** | 180 |
| **Show Attended Group? :** | No |
| **Edit Attended Group? :** | No |
| **Break time :** | 0 |
| **Group Marks :** | 300 |

## Mathematics Section A

| | |
|---|---|
| **Section Number :** | 1 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 1 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let **f :** $\mathbb{R} \to \mathbb{R}$ be a polynomial function of degree four having extreme values at **x = 4** and **x = 5**. If $\lim\limits_{x \to 0} \frac{f(x)}{x^2} = 5$, then **f(2)** is equal to :

**Options :**

A. 8

B. 10

C. 12

D. 14

**Question Number : 2 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let y = y(x) be the solution of the differential equation $(x^2 + 1)y' - 2xy = (x^4 + 2x^2 + 1)\cos x$,

$y(0) = 1$. Then $\int_{-3}^{3} y(x)\,dx$ is :

**Options :**

A. 36

B. 24

C. 18

D. 30

**Question Number : 3 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the locus of $z \in \mathbb{C}$, such that $\mathrm{Re}\left(\frac{z-1}{2z+i}\right) + \mathrm{Re}\left(\frac{z-1}{2z-i}\right) = 2$, is a circle of radius r and center $(a, b)$, then $\frac{15ab}{r^2}$ is equal to :

**Options :**

A. 16

B. 24

C. 12

D. 18

**Question Number : 4 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $a_n$ be the $n^{th}$ term of an A.P. If $S_n = a_1 + a_2 + a_3 + \ \ldots \ + a_n = 700$, $a_6 = 7$ and $S_7 = 7$, then $a_n$ is equal to :

**Options :**

A. 65

B. 56

C. 70

D. 64

**Question Number : 5 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A bag contains 19 unbiased coins and one coin with head on both sides. One coin drawn at random is tossed and head turns up. If the probability that the drawn coin was unbiased, is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $n^2 - m^2$ is equal to :

**Options :**

A. 64

B. 80

C. 60

D. 72

**Question Number : 6 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2},\ x \neq 1, 2,$ is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to :

**Options :**

A. 188

B. 192

C. 190

D. 194

**Question Number : 7 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The number of real roots of the equation $x|x-2| + 3|x-3| + 1 = 0$ is :

**Options :**

A. 4

B. 3

C. 2

D. 1

**Question Number : 8 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the length of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be 10. If its eccentricity is the minimum value of the function $f(t) = t^2 + t + \frac{11}{12}, t \in \mathbb{R}$, then $a^2 + b^2$ is equal to :

**Options :**

A. 115

B. 120

C. 125

D. 126

**Question Number : 9 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $e_1$ and $e_2$ be the eccentricities of the ellipse $\frac{x^2}{b^2}+\frac{y^2}{25}=1$ and the hyperbola $\frac{x^2}{16}-\frac{y^2}{b^2}=1$, respectively. If $b < 5$ and $e_1e_2 = 1$, then the eccentricity of the ellipse having its axes along the coordinate axes and passing through all four foci (two of the ellipse and two of the hyperbola) is :

**Options :**

A. $\frac{4}{5}$

B. $\frac{3}{5}$

C. $\frac{\sqrt{7}}{4}$

D. $\frac{\sqrt{3}}{2}$

**Question Number : 10 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the equation of the line passing through the point $\left(0,-\frac{1}{2},0\right)$ and perpendicular to the lines $\vec{r}=\lambda\left(\hat{i}+a\hat{j}+b\hat{k}\right)$ and $\vec{r}=\left(\hat{i}-\hat{j}-6\hat{k}\right)+\mu\left(-b\hat{i}+a\hat{j}+5\hat{k}\right)$ is $\frac{x-1}{-2}=\frac{y+4}{d}=\frac{z-c}{-4}$, then $a+b+c+d$ is equal to :

**Options :**

A. 13

B. 14

C. 12

D. 10

**Question Number : 11 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is :

**Options :**

A. 757

B. 755

C. 750

D. 760

**Question Number : 12 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the area of the region $\{(x, y) : 1 + x^2 \leq y \leq \min\{x + 7, 11 - 3x\}\}$ is $A$, then $3A$ is equal to :

**Options :**

A. 50

B. 46

C. 49

D. 47

**Question Number : 13 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The number of solutions of the equation

$\cos 2\theta \cos\frac{\theta}{2} + \cos\frac{5\theta}{2} = 2\cos^3\frac{5\theta}{2}$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is :

**Options :**

A. 5

B. 7

C. 6

D. 9

**Question Number : 14 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the lines $\mathbf{L_1}$: x - 1 = y - 2 = z and $\mathbf{L_2}$: x - 2 = y = z - 1. Let the feet of the perpendiculars from the point P(5, 1, -3) on the lines $\mathbf{L_1}$ and $\mathbf{L_2}$ be Q and R respectively. If the area of the triangle PQR is A, then $4A^2$ is equal to :

**Options :**

A. 151

B. 147

C. 139

D. 143

**Question Number : 15 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $\mathbf{A} = \{ (\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : |\alpha - 1| \leq 4 \text{ and } |\beta - 5| \leq 6 \}$

and $\mathbf{B} = \{ (\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : 16(\alpha - 2)^2 + 9(\beta - 6)^2 \leq 144 \}$.

Then

**Options :**

A. $A \subset B$

B. $B \subset A$

C. neither $A \subset B$ nor $B \subset A$

D. $A \cup B = \{(x, y) : -4 \leqslant x \leqslant 4, -1 \leqslant y \leqslant 11\}$

**Question Number : 16 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that

$\frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \sqrt{2} + 1$. Then $\frac{|\vec{a} + \vec{b}|^2}{|\vec{a}|^2}$ is :

**Options :**

A. $2 + \sqrt{2}$

B. $2 + 4\sqrt{2}$

C. $4 + 2\sqrt{2}$

D. $1 + \sqrt{2}$

**Question Number : 17 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let p be the number of all triangles that can be formed by joining the vertices of a regular polygon P of n sides and q be the number of all quadrilaterals that can be formed by joining the vertices of P. If p + q = 126, then the eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{n} = 1$ is :

**Options :**

A. $\frac{1}{\sqrt{2}}$

B. $\frac{1}{2}$

C. $\frac{\sqrt{7}}{4}$

D. $\frac{3}{4}$

**Question Number : 18 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let a random variable X take values 0, 1, 2, 3 with P(X=0)=P(X=1)=p, P(X=2)=P(X=3) and $E(X^2)=2E(X)$. Then the value of $8p-1$ is :

**Options :**

A. 2

B. 0

C. 3

D. 1

**Question Number : 19 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Let the system of equations

$x + 5y - z = 1$

$4x + 3y - 3z = 7$

$24x + y + \lambda z = \mu$

$\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system,

if x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is :

**Options :**

A. 4

B. 5

C. 3

D. 6

**Question Number : 20 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

If the orthocenter of the triangle formed by the lines $y = x + 1$, $y = 4x - 8$ and $y = mx + c$ is at (3, -1), then $m - c$ is :

**Options :**

A. 0

B. 2

C. -2

D. 4

## Mathematics Section B

| | |
|---|---|
| **Section Number :** | 2 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 21 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The sum of the series $2\times1\times{}^{20}C_4-3\times2\times{}^{20}C_5+4\times3\times{}^{20}C_6-5\times4\times{}^{20}C_7+\cdots\cdots+18\times17\times{}^{20}C_{20}$, is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 22 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If the function $f(x)=\frac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$ is continuous at $x=0$, then $f(0)$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 23 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

If $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3x^{-24}+x^{-26}}\right)dx=-\frac{\alpha}{3(\alpha+1)}\left(3x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+C, x>0, (\alpha,\beta,\gamma\in\mathbf{Z})$, where C is the constant of integration, then $\alpha+\beta+\gamma$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 24 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Let the lengths of the transverse and conjugate axes of a hyperbola in standard form be $2a$ and $2b$, respectively, and one focus and the corresponding directrix of this hyperbola be $(-5,0)$ and $5x+9=0$, respectively. If the product of the focal distances of a point $(\alpha,2\sqrt{5})$ on the hyperbola is $p$, then $4p$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 25 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

For $t > -1$, let $\alpha_t$ and $\beta_t$ be the roots of the equation

$$\left((t+2)^{1/7}-1\right)x^2+\left((t+2)^{1/6}-1\right)x+\left((t+2)^{1/21}-1\right)=0. \text{ If } \lim_{t\to -1^+}\alpha_t = a \text{ and } \lim_{t\to -1^+}\beta_t = b,$$

then $72(a+b)^2$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

## Physics Section A

| | |
|---|---|
| **Section Number :** | 3 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 26 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equal to that of:

($\mu_0$ = Vacuum permeability and $\epsilon_0$ = Vacuum permittivity)

**Options :**

A. Voltage

B. Inductance

C. Resistance

D. Capacitance

**Question Number : 27 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A photoemissive substance is illuminated with a radiation of wavelength $\lambda_i$ so that it releases electrons with de-Broglie wavelength $\lambda_e$. The longest wavelength of radiation that can emit photoelectron is $\lambda_o$. Expression for de-Broglie wavelength is given by:

(m: mass of the electron, h: Planck's constant and c: speed of light)

**Options :**

A. $\lambda_e = \frac{\sqrt{h\lambda_i}}{\sqrt{2mc}}$

B. $\lambda_e = \frac{h}{\sqrt{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_o}\right)}}$

C. $\lambda_e = \sqrt{\frac{h}{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_o}\right)}}$

D. $\lambda_e = \sqrt{\frac{h\lambda_0}{2mc}}$

**Question Number : 28 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A helicopter flying horizontally with a speed of 360 km/h at an altitude of 2 km, drops an object at an instant. The object hits the ground at a point O, 20 s after it is dropped. Displacement of 'O' from the position of helicopter where the object was released is :

(use acceleration due to gravity g = 10 m/s$^2$ and neglect air resistance)

**Options :**

A. 7.2 km

B. $2\sqrt{5}$ km

C. $2\sqrt{2}$ km

D. 4 km

**Question Number : 29 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

**Assertion (A):** Refractive index of glass is higher than that of air.

**Reason (R):** Optical density of a medium is directly proportionate to its mass density which results in a proportionate refractive index.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

**Options :**

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

B. (A) is correct but (R) is not correct

C. (A) is not correct but (R) is correct

D. Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

**Question Number : 30 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Which one of the following forces cannot be expressed in terms of potential energy?

**Options :**

A. Frictional force

B. Coulomb's force

C. Restoring force

D. Gravitational force

**Question Number : 31 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match **List - I** with **List - II**.

| **List - I** | **List - II** |
|---|---|
| (A) Mass density | (I) $[ML^2T^{-3}]$ |
| (B) Impulse | (II) $[MLT^{-1}]$ |
| (C) Power | (III) $[ML^2T^0]$ |
| (D) Moment of inertia | (IV) $[ML^{-3}T^0]$ |

Choose the **correct** answer from the options given below :

**Options :**

A. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

B. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

D. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

**Question Number : 32 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The helium and argon are put in the flask at the same room temperature (300 K). The ratio of average kinetic energies (per molecule) of helium and argon is:

(Give: Molar mass of helium = 4 g/mol, Molar mass of argon = 40 g/mol)

**Options :**

A. $1 : \sqrt{10}$

B. $10 : 1$

C. $1 : 10$

D. $1 : 1$

**Question Number : 33 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A):** The radius vector from the Sun to a planet sweeps out equal areas in equal intervals of time and thus areal velocity of planet is constant.

**Reason (R):** For a central force field the angular momentum is a constant.

In the light of the above statements, choose the most appropriate answer from the options given below:

**Options :**

A. (A) is not correct but (R) is correct

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C. Both (A) and (R) are correct and (R) is the correct explanation of (A)

D. (A) is correct but (R) is not correct

**Question Number : 34 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The unit of $\sqrt{\frac{2I}{\varepsilon_0 c}}$ is :

(I = intensity of an electromagnetic wave, c = speed of light)

**Options :**

A. Vm

B. $NC^{-1}$

C. NC

D. Nm

**Question Number : 35 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A dipole with two electric charges of 2 μC magnitude each, with separation distance 0.5 μm, is placed between the plates of a capacitor such that its axis is parallel to an electric field established between the plates when a potential difference of 5 V is applied. Separation between the plates is 0.5 mm. If the dipole is rotated by 30° from the axis, it tends to realign in the direction due to a torque. The value of torque is:

**Options :**

A. $2.5\times10^{-9}$ Nm

B. $2.5\times10^{-12}$ Nm

C. $5\times10^{-3}$ Nm

D. $5\times10^{-9}$ Nm

**Question Number : 36 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

**Assertion (A) :** The outer body of an aircraft is made of metal which protects persons sitting inside from lightning strikes.

**Reason (R) :** The electric field inside the cavity enclosed by a conductor is zero.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

B. (A) is correct but (R) is not correct

C. Both (A) and (R) are correct and (R) is the correct explanation of (A)

D. (A) is not correct but (R) is correct

**Question Number : 37 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A mirror is used to produce an image with magnification of $\frac{1}{4}$. If the distance between object and its image is 40 cm, then the focal length of the mirror is _____.

**Options :**

A. 10 cm

B. 12.7 cm

C. 10.7 cm

D. 15 cm

**Question Number : 38 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** The density of the copper ($^{64}_{29}$Cu) nucleus is greater than that of the carbon ($^{12}_{6}$C) nucleus.

**Reason (R) :** The nucleus of mass number A has a radius proportional to $A^{1/3}$.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

**Options :**

A. (A) is correct but (R) is not correct

B. Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

C. (A) is not correct but (R) is correct

D. Both (A) and (R) are correct and (R) **is** the correct explanation of (A)

**Question Number : 39 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A capillary tube of radius 0.1 mm is partly dipped in water (surface tension 70 dyn/cm and glass water contact angle $\approx 0°$) with 30° inclined with the vertical. The length of water risen in the capillary is _____ cm. (Take $g = 9.8\ m/s^2$)

**Options :**

A. $\frac{82}{5}$

B. $\frac{68}{5}$

C. $\frac{57}{2}$

D. $\frac{71}{5}$

**Question Number : 40 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

**Assertion (A):** Magnetic monopoles do not exist.

**Reason (R):** Magnetic field lines are continuous and form closed loops.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

**Options :**

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

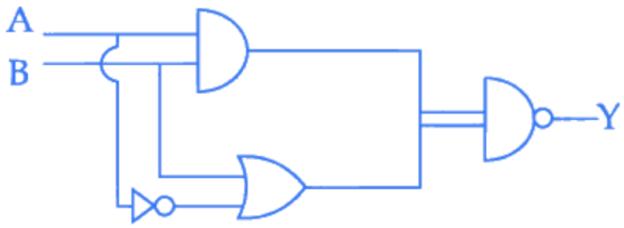
B. (A) is not correct but (R) is correct

C. (A) is correct but (R) is not correct

D. Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

**Question Number : 41 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Consider the following logic circuit.




The output is Y = 0 when :

**Options :**

A. A = 1 and B = 0

B. A = 0 and B = 0

C. A = 1 and B = 1

D. A = 0 and B = 1

**Question Number : 42 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The equation of a wave travelling on a string is $y = \sin[20\pi x + 10\pi t]$, where x and t are distance and time in SI units. The minimum distance between two points having the same oscillating speed is :

**Options :**

A. 10 cm

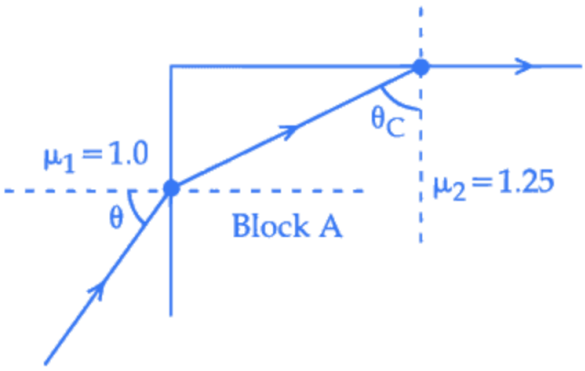
B. 2.5 cm

C. 20 cm

D. 5.0 cm

**Question Number : 43 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A transparent block A having refractive index $\mu = 1.25$ is surrounded by another medium of refractive index $\mu = 1.0$ as shown in the figure. A light ray is incident on the flat face of the block with incident angle $\theta$ as shown in the figure. What is the maximum value of $\theta$ for which light suffers total internal reflection at the top surface of the block?




**Options :**

A. $\tan^{-1}(4/3)$

B. $\sin^{-1}(3/4)$

C. $\tan^{-1}(3/4)$

D. $\cos^{-1}(3/4)$

**Question Number : 44 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match **List - I** with **List - II**.

| **List - I** | **List - II** |
|---|---|
| (A) Isothermal | (I) $\Delta W$ (work done) = 0 |
| (B) Adiabatic | (II) $\Delta Q$ (supplied heat) = 0 |
| (C) Isobaric | (III) $\Delta U$ (change in internal energy) $\neq 0$ |
| (D) Isochoric | (IV) $\Delta U = 0$ |

Choose the **correct** answer from the options given below :

**Options :**

A. (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

B. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

C. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

D. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

**Question Number : 45 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

An object with mass 500 g moves along x-axis with speed $v = 4\sqrt{x}$ m/s. The force acting on the object is :

**Options :**

A. 8 N

B. 4 N

C. 5 N

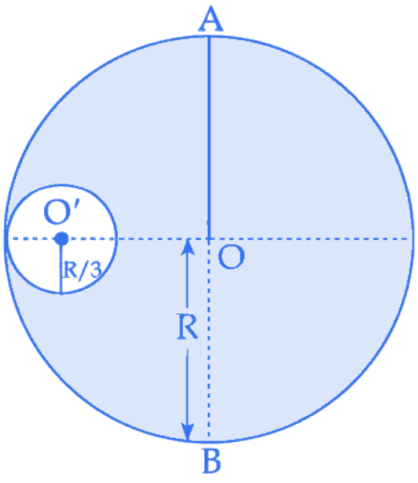
D. 6 N

## Physics Section B

| | |
|---|---|
| **Section Number :** | 4 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 46 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

M and R be the mass and radius of a disc. A small disc of radius R/3 is removed from the bigger disc as shown in figure. The moment of inertia of remaining part of bigger disc about an axis *AB* passing through the centre *O* and perpendicular to the plane of disc is $\frac{4}{x}MR^2$. The value of $x$ is _____.




Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 47 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Two cylindrical rods A and B made of different materials, are joined in a straight line. The ratios of lengths, radii and thermal conductivites of these rods are: $\frac{L_A}{L_B} = \frac{1}{2}, \frac{r_A}{r_B} = 2$ and $\frac{K_A}{K_B} = \frac{1}{2}$. The free ends of rods A and B are maintained at 400 K , 200 K , respectively. The temperature of rods interface is _____ *K*, when equilibrium is established.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 48 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The electric field in a region is given by $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3$ N/C. The flux of the field through a rectangular surface parallel to $x - z$ plane is 6.0 $Nm^2C^{-1}$. The area of the surface is _____ $cm^2$.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 49 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

An inductor of reactance 100Ω, a capacitor of reactance 50Ω, and a resistor of resistance 50Ω are connected in series with an AC source of 10 V, 50 Hz. Average power dissipated by the circuit is _____ W.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 50 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

A parallel plate capacitor has charge $5 \times 10^{-6}$C. A dielectric slab is inserted between the plates and almost fills the space between the plates. If the induced charge on one face of the slab is $4 \times 10^{-6}$C then the dielectric constant of the slab is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

## Chemistry Section A

| | |
|---|---|
| **Section Number :** | 5 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 20 |
| **Number of Questions to be attempted :** | 20 |
| **Section Marks :** | 80 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 51 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match **List - I** with **List - II**.

| **List - I (Complex)** | **List - II (Primary valency and Secondary valency)** |
|---|---|
| (A) $[Co(en)_2Cl_2]Cl$ | (I) 3, 6 |
| (B) $[Pt(NH_3)_2Cl(NO_2)]$ | (II) 3, 4 |
| (C) Hg $[Co(SCN)_4]$ | (III) 2, 6 |
| (D) $[Mg\ (EDTA)]^{2-}$ | (IV) 2, 4 |

Choose the **correct** answer from the options given below :

**Options :**

A. (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

B. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C. (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

D. (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

**Question Number : 52 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The correct statements from the following are :

(A) $Tl^{3+}$ is a powerful oxidising agent

(B) $Al^{3+}$ does not get reduced easily

(C) Both $Al^{3+}$ and $Tl^{3+}$ are very stable in solution

(D) $Tl^{1+}$ is more stable than $Tl^{3+}$

(E) $Al^{3+}$ and $Tl^{1+}$ are highly stable

Choose the **correct** answer from the options given below :

**Options :**

A. (A), (B), (D) and (E) only

B. (A), (C) and (D) only

C. (A), (B), (C) and (E)

D. (B), (D) and (E) only

**Question Number : 53 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

**Statement (I) :** On hydrolysis, oligo peptides give rise to fewer number of α-amino acids while proteins give rise to a large number of β-amino acids.

**Statement (II) :** Natural proteins are denatured by acids which convert the water soluble form of fibrous proteins to their water insoluble form.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

**Options :**

A. **Statement I** is incorrect but **Statement II** is correct

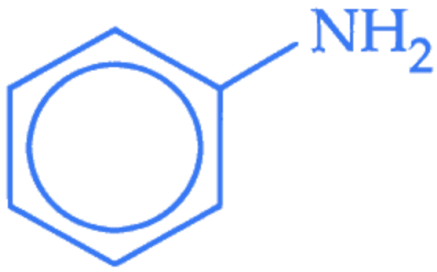
B. Both **Statement I** and **Statement II** are incorrect

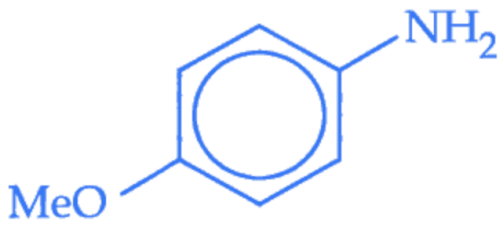
C. Both **Statement I** and **Statement II** are correct

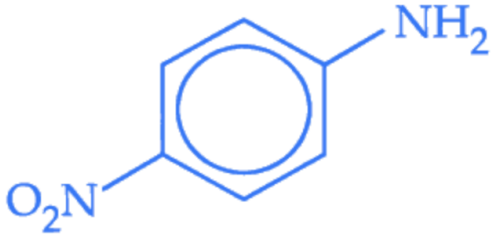
D. **Statement I** is correct but **Statement II** is incorrect

**Question Number : 54 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The descending order of basicity of the following amines is :

(A) 


(B) 


(C) 


(D) $CH_3NH_2$

(E) $(CH_3)_2NH$

Choose the **correct** answer from the options given below :

**Options :**

A. $E > A > D > C > B$

B. $E > D > A > B > C$

C. $E > D > B > A > C$

D. $B > E > D > A > C$

**Question Number : 55 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

In $SO_2$, $NO_2^-$ and $N_3^-$, the hybridizations at the central atom are respectively :

**Options :**

A. $sp^2$, $sp^2$ and $sp^2$

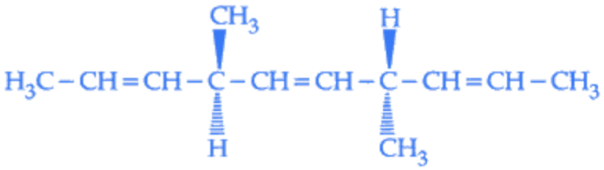
B. sp, $sp^2$ and sp

C. $sp^2$, sp and sp

D. $sp^2$, $sp^2$ and sp

**Question Number : 56 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The number of optically active products obtained from the complete ozonolysis of the given compound is :



**Options :**

A. 4

B. 0

C. 2

D. 1

**Question Number : 57 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

A mixture of 1 g each of chlorobenzene, aniline, and benzoic acid is dissolved in 50 mL ethyl acetate and placed in a separating funnel. 5 M NaOH (30 mL) was added in the same funnel. The funnel was shaken vigorously and then kept aside. The ethyl acetate layer in the funnel contains :

**Options :**

A. chlorobenzene and aniline

B. benzoic acid and aniline

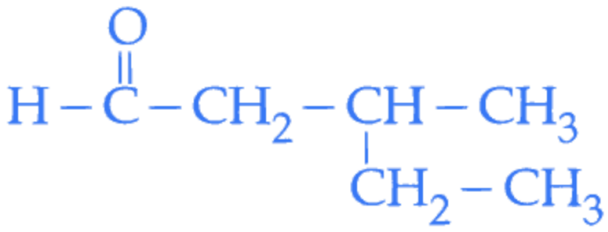
C. benzoic acid

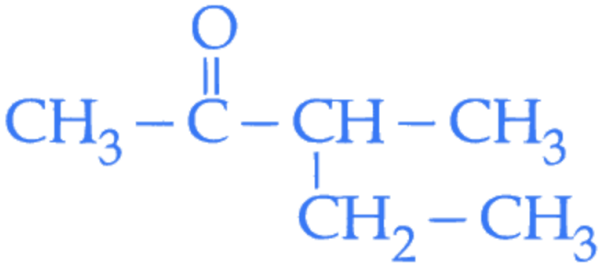
D. benzoic acid and chlorobenzene

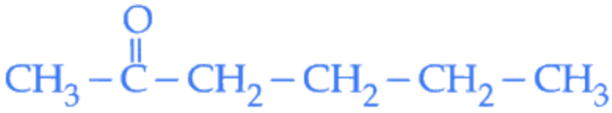
**Question Number : 58 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

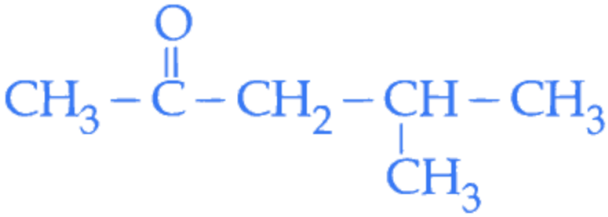
" P " is an optically active compound with molecular formula $C_6H_{12}O$. When ' P ' is treated with 2, 4-dinitrophenylhydrazine, it gives a positive test. However, in presence of Tollens reagent, " P " gives a negative test. Predict the structure of " P ".

**Options :**

A. 

B. 

C. 

D. 

**Question Number : 59 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

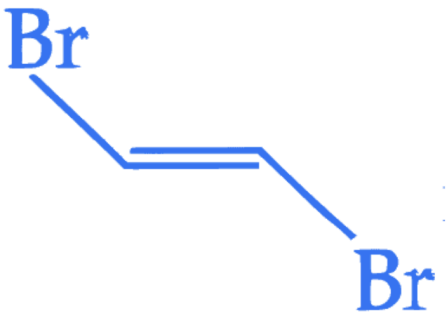
**Statement (I)** :




is more polar than




**Statement (II)** : Boiling point of




is lower than




but it is more polar than




.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

**Options :**

A. **Both Statement I and Statement II are correct**

B. **Statement I is incorrect but Statement II is correct**

C. **Both Statement I and Statement II are incorrect**

D. **Statement I is correct but Statement II is incorrect**

**Question Number : 60 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Choose the incorrect trend in the atomic radii (r) of the elements.

**Options :**

A. $r_{Rb} < r_{Cs}$

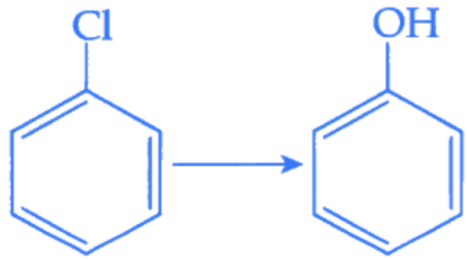
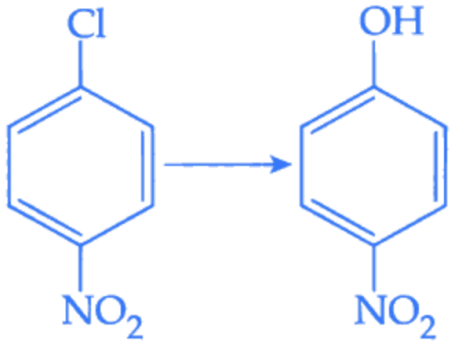
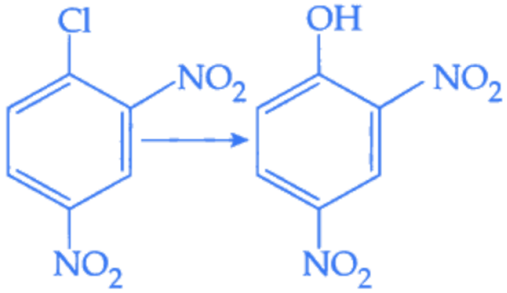
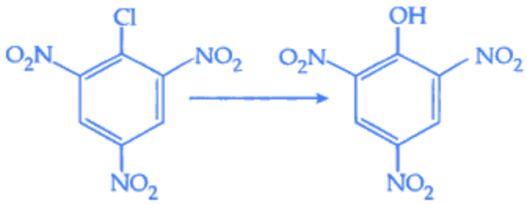
B. $r_{At} < r_{Cs}$

C. $r_{Br} < r_{K}$

D. $r_{Mg} < r_{Al}$

**Question Number : 61 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match **List - I** with **List - II**.

| **List - I (Conversion)** | **List - II (Reagents, Conditions used)** |
|---|---|
| (A)  | (I) Warm $H_2O$ |
| (B)  | (II) (a) NaOH, 368K; (b) $H_3O^+$ |
| (C)  | (III) (a) NaOH, 443K; (b) $H_3O^+$ |
| (D)  | (IV) (a) NaOH, 623K, 300 atm; (b) $H_3O^+$ |

Choose the **correct** answer from the options given below :

**Options :**

A. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

B. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

C. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

D. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

**Question Number : 62 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Liquid A and B form an ideal solution. The vapour pressures of pure liquids A and B are 350 and 750 mm Hg respectively at the same temperature. If $x_A$ and $x_B$ are the mole fraction of A and B in solution while $y_A$ and $y_B$ are the mole fraction of A and B in vapour phase, then,

**Options :**

A. $(x_A - y_A) < (x_B - y_B)$

B. $\frac{x_A}{x_B} = \frac{y_A}{y_B}$

C. $\frac{x_A}{x_B} < \frac{y_A}{y_B}$

D. $\frac{x_A}{x_B} > \frac{y_A}{y_B}$

**Question Number : 63 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Match **List - I** with **List - II**.

| **List - I** | **List - II** |
|---|---|
| (A) Solution of chloroform and acetone | (I) Minimum boiling azeotrope |
| (B) Solution of ethanol and water | (II) Dimerizes |
| (C) Solution of benzene and toluene | (III) Maximum boiling azeotrope |
| (D) Solution of acetic acid in benzene | (IV) $\Delta V_{mix} = 0$ |

**Choose the correct answer from the options given below :**

**Options :**

A. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

C. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

**Question Number : 64 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The hydration energies of $K^+$ and $Cl^-$ are $-x$ and $-y$ kJ/mol respectively. If the lattice energy of KCl is $-z$ kJ/mol, then the heat of solution of KCl is :

**Options :**

A. $x + y + z$

B. $z-(x+y)$

C. $-z-(x+y)$

D. $x-y-z$

**Question Number : 65 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The number of unpaired electrons responsible for the paramagnetic nature of the following complex species are respectively :

$[Fe(CN)_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$

**Options :**

A. 1, 5, 4, 2

B. 1, 4, 4, 2

C. 1, 5, 5, 2

D. 1, 1, 4, 2

**Question Number : 66 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

$A(g) \rightarrow B(g) + C(g)$ is a first order reaction.

| **Time** | **t** | $\infty$ |
|---|---|---|
| $P_{system}$ | $P_t$ | $P_\infty$ |

The reaction was started with reactant A only. Which of the following expressions is correct for rate constant k?

**Options :**

A. $k = \frac{1}{t}\ln\frac{p_\infty}{p_t}$

B. $k = \frac{1}{t}\ln\frac{p_\infty}{2(p_\infty - p_t)}$

C. $k = \frac{1}{t}\ln\frac{2(p_\infty - p_t)}{p_t}$

D. $k = \frac{1}{t}\ln\frac{p_\infty}{(p_\infty - p_t)}$

**Question Number : 67 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

Given below are two statements :

1 M aqueous solutions of each of $Cu(NO_3)_2$, $AgNO_3$, $Hg_2(NO_3)_2$, $Mg(NO_3)_2$ are electrolysed using inert electrodes. Given: $\mathbf{E}^0_{Ag^+/Ag}$ = 0.80 V, $\mathbf{E}^0_{Hg_2^{2+}/Hg}$ = 0.79 V, $\mathbf{E}^0_{Cu^{2+}/Cu}$ = 0.24 V and $\mathbf{E}^0_{Mg^{2+}/Mg}$ = -2.37 V.

**Statement (I) :** With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu.

**Statement (II) :** Magnesium will not be deposited at the cathode instead oxygen gas will be evolved at the cathode.

In the light of the above statements, choose the most appropriate answer from the options given below :

**Options :**

A. Both **Statement I** and **Statement II** are incorrect

B. **Statement I** is incorrect but **Statement II** is correct

C. **Statement I** is correct but **Statement II** is incorrect

D. Both **Statement I** and **Statement II** are correct

**Question Number : 68 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The extra stability of half-filled subshell is due to :

(A) Symmetrical distribution of electrons

(B) Smaller coulombic repulsion energy

(C) The presence of electrons with the same spin in non-degenerate orbitals

(D) Larger exchange energy

(E) Relatively smaller shielding of electrons by one another

Indentify the correct statements :

**Options :**

A. (B), (D) and (E) only

B. (*A*), (*B*), (*D*) and (E) only

C. (B), (C) and (D) only

D. (*A*), (*B*) and (D) only

**Question Number : 69 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

'X' is the number of acidic oxides among $VO_2$, $V_2O_3$, $CrO_3$, $V_2O_5$ and $Mn_2O_7$. The primary valency of cobalt in $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$ is Y. The value of X + Y is _____.

**Options :**

A. 3

B. 4

C. 2

D. 5

**Question Number : 70 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical**

The correct statement amongst the following is :

**Options :**

A. The standard state of a pure gas is the pure gas at a pressure of 1 bar and temperature 273 K.

B. $\Delta_f H^\circ_{500}$ is zero for $O_2(g)$.

C. $\Delta_f H^\circ_{298}$ is zero for $O(g)$.

D. The term 'standard state' implies that the temperature is 0°C.

## Chemistry Section B

| | |
|---|---|
| **Section Number :** | 6 |
| **Section type :** | Online |
| **Mandatory or Optional :** | Mandatory |
| **Number of Questions :** | 5 |
| **Number of Questions to be attempted :** | 5 |
| **Section Marks :** | 20 |
| **Maximum Instruction Time :** | 0 |
| **Sub-Section Number :** | 1 |
| **Question Shuffling Allowed :** | Yes |
| **Is Section Default? :** | No |

**Question Number : 71 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

In Dumas' method 292 mg of an organic compound released 50 mL of nitrogen gas $(N_2)$ at 300 K temperature and 715 mm Hg pressure. The percentage composition of ' N ' in the organic compound is _____ % (Nearest integer) (Aqueous tension at 300 K = 15 mmHg )

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText

**Question Number : 72 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Butane reacts with oxygen to produce carbon dioxide and water following the equation given below.

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$$

If 174.0 kg of butane is mixed with 320.0 kg of $O_2$, the volume of water formed in liters is

_____. (Nearest integer)

[Given : (a) Molar mass of C, H, O are 12,1,16 g $mol^{-1}$ respectively, (b) Density of water $= 1\ g\ mL^{-1}$]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 73 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

The number of paramagnetic metal complex species among $\left[Co(NH_3)_6\right]^{3+}$, $\left[Co(C_2O_4)_3\right]^{3-}$, $\left[MnCl_6\right]^{3-}$, $\left[Mn(CN)_6\right]^{3-}$, $\left[CoF_6\right]^{3-}$, $\left[Fe(CN)_6\right]^{3-}$ and $\left[FeF_6\right]^{3-}$ with same number of unpaired electrons is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 74 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

One litre buffer solution was prepared by adding 0.10 mol each of $NH_3$ and $NH_4Cl$ in deionised water. The change in pH on addition of 0.05 mol of HCl to the above solution is _____ $\times 10^{-2}$.

(Nearest integer)

Given : $pK_b$ of $NH_3 = 4.745$ and $\log_{10} 3 = 0.477$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

**Question Number : 75 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript**

Identify the structure of the final product (D) in the following sequence of the reactions :




Total number of $sp^2$ hybridised carbon atoms in product D is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText